SIMATS ENGINEERING

ASSESSMENT TOOL 2

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COURSE NAME: Design and Analysis of Algorithms

COURSE CODE: CSA0613

Scenario Based Assessment

Q13. Debugging Bubble Sort (Incorrect Loop Bounds)

Scenario: A Bubble Sort implementation submitted for grading fails to sort correctly due to improper loop bounds. The error in the iteration limits must be identified, its effect explained, the code debugged step by step, the algorithm optimized by reducing redundant passes, the time complexity of both versions analyzed, and a corrected version presented.

a) Identifying the Error and Debugging It

The inner loop is written as "for j in range(n)", which lets j take every value from 0 to n-1. Inside the loop, the code accesses arr[j+1]. When j reaches its final value, n-1, the code tries to access arr[n], which is one position past the last valid index (arrays are indexed 0 to n-1).

- Incorrect bound: range(n) allows j = n-1, causing arr[j+1] = arr[n] to be accessed.
- Correct bound: the inner loop should stop at n-i-2 (i.e. range(n-i-1)), so the last comparison is always between arr[n-2] and arr[n-1].
- A second, related issue: even if the crash is patched (e.g. range(n-1)), the outer loop still runs n full passes even after the array becomes sorted, wasting comparisons on later passes.

Effect of the error: Because the loop bound is one too large, arr[j+1] is evaluated with j+1 = n on the last iteration of every pass. Since valid indices only go up to n-1, Python raises an IndexError: list index out of range and the program terminates before sorting completes. The array is left partially processed, and no sorted result is ever returned.

Source Code (Buggy):

```
def bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        for j in range(n):
            if arr[j] > arr[j+1]:   # ERROR
                arr[j], arr[j+1] = arr[j+1], arr[j]

arr = [5, 2, 9, 1, 5, 6]
bubble_sort(arr)
print("Sorted array:", arr)
```

Fig. 1 — Original buggy source code (bubble_sort_buggy.py)

Output (Buggy):

```
Traceback (most recent call last):
  File "bubble_sort_buggy.py", line 9, in <module>
    bubble_sort(arr)
  File "bubble_sort_buggy.py", line 5, in bubble_sort
    if arr[j] > arr[j+1]:   # ERROR
               ~~~^^^^^
IndexError: list index out of range
```

Fig. 2 — Program output: IndexError raised when running the buggy code

Step-by-step trace using the sample array [5, 2, 9, 1, 5, 6] (n = 6):

- i = 0, j goes 0..5 (range(6)). At j = 4: compare arr[4] and arr[5] — valid.
- At j = 5: the code compares arr[5] and arr[6] — arr[6] does not exist, so Python raises IndexError immediately, before any full pass or sort can complete.
- Root cause confirmed: the inner loop's range must exclude the last index that would push j+1 out of bounds.

Debugging steps applied to fix it:

- Step 1 — Change range(n) to range(n - i - 1) in the inner loop. This guarantees j+1 never exceeds n-1.
- Step 2 — Change the outer loop from range(n) to range(n - 1), since after n-1 passes the array is guaranteed sorted (the last element needs no pass of its own).
- Step 3 — Re-run with the same test array to confirm no exceptions occur and the output is correctly sorted in ascending order.
- Step 4 — Add a 'swapped' flag to detect when a pass makes zero swaps, meaning the array is already sorted, and exit early.

b) Optimization — Reducing Redundant Passes

Two optimizations are applied on top of the bug fix:

- Shrinking inner range (n - i - 1): after each pass, the largest unsorted element 'bubbles' to its correct position at the end. There is no need to re-compare the already-sorted tail, so each successive pass is made one comparison shorter.
- Early-exit 'swapped' flag: if a complete pass makes no swaps, the array is already sorted, and the algorithm stops immediately instead of running all n-1 passes regardless of input. This turns the best case from $O(n^2)$ into O(n).

Corrected & Optimized Source Code:

```
def bubble_sort(arr):
    n = len(arr)
    comparisons = 0
    swaps = 0
    for i in range(n - 1):
        swapped = False
        for j in range(n - i - 1):
            comparisons += 1
            if arr[j] > arr[j+1]:
                arr[j], arr[j+1] = arr[j+1], arr[j]
                swaps += 1
                swapped = True
        if not swapped:
            break
```

```
    return arr, comparisons, swaps

arr = [5, 2, 9, 1, 5, 6]
sorted_arr, comparisons, swaps = bubble_sort(arr)
print("Sorted array:", sorted_arr)
print("Comparisons:", comparisons)
print("Swaps:", swaps)
```

Fig. 3 — Corrected and optimized source code (bubble_sort_corrected.py)

Output (Corrected):

```
Sorted array: [1, 2, 5, 5, 6, 9]
Comparisons: 14
Swaps: 7

=== Code Execution Successful ===
```

Fig. 4 — Program output: correct sorted result, comparison/swap counts, and best-case early exit

c) Time Complexity Analysis

Case	Buggy Version	Corrected & Optimized Version
Best Case	Crashes — IndexError (never reaches a valid result)	O(n) — already-sorted array, one pass then early exit via 'swapped' flag
Average Case	Crashes — IndexError on the last comparison of every pass	$O(n^2)$ — typical random ordering, no early exit
Worst Case	Crashes — IndexError (reverse-sorted or any input, since j+1 always overshoots at j = n-1)	$O(n^2)$ — reverse-sorted array, full n-1 passes required
Space Complexity	N/A	O(1) — in-place sort, only constant extra variables used

Interpretation: The bug is not a performance problem — it is a correctness problem, since the original code never completes and always crashes with IndexError regardless of input. Once corrected, the algorithm has the standard Bubble Sort complexity of $O(n^2)$ in the average and worst cases (nested comparisons over the unsorted portion) and O(1) space (sorting is done in place). The 'swapped' flag optimization improves only the best case, bringing it down to O(n), which matters when the input is already sorted or nearly sorted — a common case in real-world data such as small, mostly-ordered lists.

Report

Aim

To debug a Bubble Sort implementation containing incorrect loop bounds, identify and fix the resulting IndexError, optimize it to reduce redundant passes, and analyze the time and space complexity of the corrected version.

Algorithm

- Read the input array and determine its length, n.
- For each outer pass i from 0 to n-2, reset a 'swapped' flag to False.
- For each inner index j from 0 to n-i-2, compare arr[j] and arr[j+1].
- If arr[j] > arr[j+1], swap them and set 'swapped' to True.
- If no swaps occurred during a full pass, break early — the array is already sorted.
- Repeat until all passes are complete or an early exit occurs.

Time Complexity

- Best Case: O(n) — already-sorted input, one pass with early exit.
- Average Case: O(n^2) — typical random ordering, no early exit.
- Worst Case: O(n^2) — reverse-sorted input, full n-1 passes required.

Space Complexity

O(1) — in-place sort, only constant extra variables (i, j, swapped, comparisons, swaps) are used.

Conclusion

The original implementation's inner loop used range(n) instead of range(n - i - 1), causing arr[j+1] to be indexed one position beyond the array's bounds and crashing with an IndexError on every pass. The corrected version fixes the loop bounds so the algorithm terminates correctly, and adds an early-exit 'swapped' flag so that already-sorted or nearly-sorted arrays are processed in O(n) instead of always paying the full O(n^2) cost. The corrected and optimized version was tested and verified to produce the correct sorted output, as shown in Fig. 3 and Fig. 4 above.

Assessment Rubric Reference

The table below reproduces the grading rubric used to evaluate responses to this task.

Criteria	Weightage	Exemplary	Proficient	Developing	Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification